

**JIMMA UNIVERSITY
INSTITUTE OF
TECHNOLOGY
DEPARTMENT OF CIVIL
ENGINEERING**

Exit

Exam Model 1_2024

** Indicates required question*

Personal Information

1. Full Name *

2. ID Number *

3. *
SECTION

Skip to question 4

Exam Questions

4. Type of specification focuses on the physical and or chemical properties that can also * 1 point be crosschecked by tests and for others the performance characteristics.

Mark only one oval.

- ☐ A. Workmanship Specifications
- ☐ B. Material Specifications
- ☐ C. Performance Specifications
- ☐ D. None

5. which of the following is not true about “Out – to - Out and In – to – In” takeoff Method * 1 point

Mark only one oval.

- ☐ A. Long Wall (Out – to - Out) = Inner Length + 2 Times Thickness of the Wall
- ☐ B. Short Wall (In – to – In) = Inner Length
- ☐ C. Short Wall (In – to – In) = outer Length - 2 Times Thickness of the Wall
- ☐ D. None

6. Which of the following is not true about the fluctuation of amount of water in soil? * 1 point

Mark only one oval.

- ☐ a) Strength decreases as water content increases
- ☐ b) Soils swell-up when water content increases
- ☐ c) Fine-grained soils at very high water content possess properties similar to liquids
- ☐ d) None of the above

7. The hydraulic gradient is equal to the _____ * 1 point

Mark only one oval.

- ☐ a) Hydraulic head
- ☐ b) velocity head
- ☐ c) hydraulic head per unit distance
- ☐ d) pressure head

8. Determine the quantity of pit excavation for square footing pad with dimensions 120cmx120cmx50cm and the total depth of foundation is 250cm excluding 5cm lean concrete below the concrete pad. * 1 point

Mark only one oval.

- ☐ A. 12.34m³
- ☐ B. 5.74m³
- ☐ C. 7.37m³
- ☐ D. 3.67m³

9. If a fully saturated soil mass has a water content of 100%, then its void ratio is _____

* 1 point

Mark only one oval.

- ☐ a) less than the specific gravity
- ☐ b) greater than the specific gravity
- ☐ c) equal to specific gravity
- ☐ d) does not depend on specific gravity

10. The stresses due to self-weight of the soil are known as _____ *

1 point

Mark only one oval.

- ☐ a) Geostatic stresses
- ☐ b) boundary stresses
- ☐ c) external stresses
- ☐ d) boundary strain

11. What will happen to the vertical effective stress when the seepage occurs in upward direction? 1 point

Mark only one oval.

- ☐ a) Increase
- ☐ b) Decrease
- ☐ c) Remain the same
- ☐ d) None

12. Calculate the quantity of block work for HCB wall dimensioned with 5m*2.80m and having door with a size of 0.9m*2.10m. * 1 point

Mark only one oval.

- ☐ A. 14m²
- ☐ B. 13.62m³
- ☐ C. 12.11m²
- ☐ D. None

13. The compaction reduces _____ *

5 / 35

Mark only one oval.

- ☐ Option 1a) Strength
- ☐ b) stability
- ☐ c) air voids
- ☐ d) dry density



14. Which of the following is correct about Indirect costs of construction *

1 point

Mark only one oval.

- ☐ A. All costs which do not become a final part of the installation or constructed facility,
- ☐ B. Which are required for the orderly completion of the installation; costs incurred indirectly for the accomplishment of the project.
- ☐ C. Which can not be directly booked under a specific activity in a construction project but required to keep the whole project operational.
- ☐ D. Indirect costs= Head office overhead costs - Site overhead costs

15. Which of the following can not be considered as basic element of contract *

1 point

Mark only one oval.

- ☐ A. Mutual Agreement
- ☐ B. Legal Objective
- ☐ C. Ethical Consideration
- ☐ D. Legal Capacity of the parties

16. According to Eurocode/EBCS code, what is the primary purpose of inelastic moment redistribution in RC beams?

* 1 point

Mark only one oval.

- ☐ A. To increase the beam's overall stiffness
- ☐ B. To redistribute internal forces and ensure ductile behavior
- ☐ C. To minimize the deflection of the beam under load
- ☐ D. To reduce the overall cost of the beam

17. If an RC column is subjected to bi-axial loading with a moment along the x-axis twice that along the y-axis, what would be your recommendation for the selection of the column cross-section (width x height) that is both economical and optimum, assuming the width (b) is parallel to the x-axis and the height (h) is parallel to the y-axis? * 1 point

Mark only one oval.

- ☐ A. Width b should be greater than height h
- ☐ B. Width b should be less than height h
- ☐ C. Width b should be equal to height h
- ☐ D. Difficult to estimate

18. What is the primary purpose of overlapping reinforcement in a two-way slab? * 1 point

Mark only one oval.

- ☐ A. To reduce the overall cost
- ☐ B. To improve the aesthetic appearance of the slab
- ☐ C. To provide continuity and develop the full strength of the reinforcement
- ☐ D. To control shear failure

19. The design of two-way slabs involves checking for deflection limits to ensure: * 1 point

Mark only one oval.

- ☐ A. Serviceability
- ☐ B. Ultimate limit state
- ☐ C. Flexural Resistance/Capacity
- ☐ D. Strength

20. What primary function guides the inclusion of drop panels in flat slab-column connections, particularly focusing on their role in enhancing structural performance at the interface between the slab and columns? * 1 point

Mark only one oval.

- ☐ A. To reduce the overall depth of the slab
- ☐ B. To increase the span-to-depth ratio
- ☐ C. To improve the stiffness and punching shear resistance at column locations
- ☐ D. To minimize deflections under live load

21. Which parameter is not typically considered in determining the span-effective depth ratios in a two-way slab? 1 point

Mark only one oval.

- ☐ A. Concrete strength
- ☐ B. Bending moment and shear force
- ☐ C. Support conditions
- ☐ D. Tension reinforcement ratio at mid-span

22. In which condition you may **not** recommend deep foundation? 1 point

Mark only one oval.

- ☐ A. When there is high thickness of compressible soil
- ☐ B. When there is a horizontal force on foundation
- ☐ C. When there is uplift pressure on foundation
- ☐ D. When there is high water level

23. Which geosynthetic material is suitable for reinforcement purpose? 1 point

Mark only one oval.

- ☐ A. geogrid
- ☐ B. geotextile
- ☐ C. geomembrane
- ☐ D. geonet

24. One of the following is not about concrete pile 1 point

Mark only one oval.

- ☐ A. High level of noise during pile driving
- ☐ B. Can be easily combined with superstructure
- ☐ C. Difficult to achieve proper cutoff
- ☐ D. Corrosion resistant
- ☐ E. None

25. What is the critical depth of pile having 0.4m diameter and 20m length constructed in homogenous sand soil? * 1 point

Mark only one oval.

- ☐ A. 4m
- ☐ B. 6m
- ☐ C. 8m
- ☐ D. 10m
- ☐ E. 12m

26. Which factor is NOT considered when calculating wind loads as per EBCS EN 1991- 1 point

Mark only one oval.

- ☐ a) Basic wind speed
- ☐ b) Exposure factor
- ☐ c) Shape factor
- ☐ d) Building weight

27. A square pile having dimension of 0.4m x 0.4m is embedded in layered saturated clay of 30m. Calculate end bearing capacity of this pile. The following are the detail of subsoil: *

Depth from ground surface (m)	Saturated unit weight (kN/m^3)	Cu (kN/m^2)
0 - 5	18	30
5- 10	18	30
10 - 30	20	100

Mark only one oval.

- ☐ A. 16 kN
- ☐ B. 144 kN
- ☐ C. 43.2 kN
- ☐ D. 320 kN

28. One of the following is **not** a requirement to select an appropriate sheet pile for your site. * 1 point

Mark only one oval.

- ☐ A. Length
- ☐ B. Shape
- ☐ C. Section Modulus
- ☐ D. Penetration Depth

29. If the dam site is situated in an earthquake zone, its design must include the Earthquake forces. But for better resistance, which pair of dams could be proposed? * 1 point

Mark only one oval.

- ☐ a. Arch and Gravity dam
- ☐ b. Rock fill and Buttress dam
- ☐ c. Earth fill and concrete Gravity dam
- ☐ d. Arch and Buttress dam



38. Which of the following conditions could lead to a negative pore water pressure in soil? * 1 point

Mark only one oval.

- ☐ A) Saturated soil under compression
- ☐ B) Dry soil
- ☐ C) Partially saturated soil experiencing capillary action
- ☐ D) Fully saturated soil undergoing rapid loading

39. In unsaturated soils, the effective stress principle is modified. The shear strength equation incorporates an additional term involving: * 1 point

Mark only one oval.

- ☐ A) Matric suction
- ☐ B) Total stress
- ☐ C) Pore water pressure
- ☐ D) Pore air pressure

40. Which aspect of soil behavior is particularly significant when considering the long-term stability of a slope in a clayey soil? * 1 point

Mark only one oval.

- ☐ A) Immediate undrained shear strength
- ☐ B) Long-term drained shear strength
- ☐ C) Rate of consolidation
- ☐ D) Rate of loading

41. What is the primary purpose of soil exploration in foundation engineering? * 1 point

Mark only one oval.

- ☐ A. To determine the color of the soil
- ☐ B. To identify the type and properties of soil
- ☐ C. To plant trees
- ☐ D. To remove topsoil

34. Which one of the following is major Mode of failure of gravity dam? *

Mark only one oval.

- ☐ a) Hydraulic, Seepage and structural failure
- ☐ b) Over topping, erosion of d/s face and piping
- ☐ c) Conduit leakage, Sloughing of downstream toe and Piping through the foundation of the dam
- ☐ d) Over turning, Sliding and overstress



35. Which method provides the most conservative estimate of bearing capacity for a granular soil? * 1 point

Mark only one oval.

- ☐ A) Terzaghi's method
- ☐ B) Meyerhof's method
- ☐ C) Vesic's method
- ☐ D) Hansen's method

36. Which of the following can increase the stability of a slope? *

1 point

Mark only one oval.

- ☐ A) Increasing the slope angle
- ☐ B) Removing vegetation
- ☐ C) Lowering the water table
- ☐ D) Increasing the weight on the slope

37. If a retaining wall moves away from the backfill soil, the lateral earth pressure against the wall:

* 1 point

Mark only one oval.

- ☐ A) Increases to at-rest earth pressure.
- ☐ B) Decreases to active earth pressure.
- ☐ C) Remains the same.
- ☐ D) Increases to passive earth pressure.

30. One of the following is not the determining factor of the type of the dam for a given * 1 point site.

Mark only one oval.

- ☐ a. Geology
- ☐ b. Available material
- ☐ c. Sediment rate
- ☐ d. Required height of dam

31. What is the most critical condition of slide of the U/s slope of embankment dam? * 1 point

Mark only one oval.

- ☐ a. The sudden drawdown of the reservoir water
- ☐ b. Reservoir full condition
- ☐ c. Steady seepage condition
- ☐ d. Sudden drawdown and reservoir full condition

32. Cracking failure of embankment dam due to frost action is _____ type of * 1 point failure

Mark only one oval.

- ☐ a. Seepage failure
- ☐ b. Structural failure
- ☐ c. Hydraulic failures
- ☐ d. Liquefaction

33. What is the maximum water pressure at the face of a gravity dam with a water * 1 point depth of 60 meters? Assuming unit weight of water is 10 KN/m³

Mark only one oval.

- ☐ a) 43200 KN/m²
- ☐ b) 1800 KN/m²
- ☐ c) 24500 KN/m²
- ☐ d) 500 KN/m²



42. What is a mat foundation best suited for? *

Mark only one oval.

- ☐ A. Light, evenly distributed loads
- ☐ B. Heavy, unevenly distributed loads
- ☐ C. Sandy soil conditions
- ☐ D. Situations with low water table

43. What is the main factor in selecting the type of foundation for a structure? *

1 point

Mark only one oval.

- ☐ A. Soil bearing capacity
- ☐ B. Color of the soil
- ☐ C. Amount of rainfall
- ☐ D. Type of vegetation

44. What is the main consideration in designing the width of a shallow foundation? *

1 point

Mark only one oval.

- ☐ A. Weather conditions
- ☐ B. Soil bearing capacity
- ☐ C. Load characteristics
- ☐ D. Depth of foundation

45. What is the primary objective of ground improvement techniques? *

1 point

Mark only one oval.

- ☐ A. To enhance the aesthetic appeal of the soil
- ☐ B. To increase the soil bearing capacity and stability
- ☐ C. To reduce soil temperature
- ☐ D. To promote plant growth

46. Soil nailing is a technique used to: *

Mark only one oval.

- ☐ A. Reinforce soil in slopes and excavations
- ☐ B. Reduce soil moisture content
- ☐ C. Improve soil fertility
- ☐ D. Measure soil resistance



47. In a singly reinforced beam, if the permissible stress in concrete reaches earlier than * 1 point that in steel, the beam section is called -----and the type of failure is-----

Mark only one oval.

- ☐ A. Under- reinforced section/ Compression failure
- ☐ B. Over- reinforced section /Compression failure
- ☐ C. Critical section/tension failure
- ☐ D. Balanced section/ tension failure

48. The maximum area of tension reinforcement in beams shall not exceed ----- of its * 1 point gross area.

Mark only one oval.

- ☐ A. 0.15%
- ☐ B. 0.26%
- ☐ C. 4%
- ☐ D. 1%

49. What will be the neutral axis depth for a singly reinforced beam whose width is 250mm and effective depth is 400 mm having a tension reinforcement of 3-20mm with grade S-400 and concrete grade of C-20, .Use equivalent rectangular stress block. * 1 point

Mark only one oval.

- ☐ A. 145mm
- ☐ B. 150mm
- ☐ C. 155mm
- ☐ D. 135mm

50. The design state deals with strength, overturning, buckling of structural members is * 1 point



Mark only one oval.

- ☐ A. Serviceability limit state
- ☐ B. Working Stress method
- ☐ C. Limit state of collapse
- ☐ D. Ultimate strength design method

51. The serviceability limit state are satisfied through the following conditions, except * 1 point

Mark only one oval.

- ☐ A. Appropriate factor of safety are used
- ☐ B. Adequate concrete cover is used
- ☐ C. Minimum depth for deflection is provided
- ☐ D. Limiting the spacing of reinforcements

52. T - beam behaves as a rectangular beam of a width equal to its flange in case * 1 point

Mark only one oval.

- ☐ A. It is lightly loaded
- ☐ B. It is heavily loaded
- ☐ C. Its natural axis remains within the web
- ☐ D. Its natural axis remains within the flange

53. A study of freeway flow at a particular site has resulted in a calibrated speed-density relationship, as follows: * 1 point

$U_s = 57.5(1 - 0.008k)$. From this relation, Find the free-flow speed and jam density?

Mark only one oval.

- ☐ a) 57.5 Km/hr, 126 veh/lane.
- ☐ b) 57.5 Km/hr, 126veh/km.
- ☐ c) 28.75 Km, hr./ 120veh/km
- ☐ d) 115 Km/hr, 120 veh/ lane

61. A vehicle travelling on dry, levelled pavement at 80 km/h had the brakes applied. * 1 po
The vehicle travelled 76.5 m before stopping. What is the coefficient of friction that has developed?

Mark only one oval.

- ☐ a) 0.2
☐ b) 0.3
☐ c) 0.33
☐ d) 0.4

62. If the cost of roadway excavation, C_e , is 20 birr/m³, cost of borrow, C_b , is 24 birr/m³, and cost of overhaul, C_{oh} , is 8 birr/m³-station, what is the economical length of overhaul? The free haul distance is 250 m and a station is 100m long. * 1 point

Mark only one oval.

- ☐ a) 400 m
☐ b) 650 m
☐ c) 550m
☐ d) 500 m

63. On a mass-haul diagram peaks indicate _____. * 1 point

Mark only one oval.

- ☐ a) a change from cut to fill
☐ b) a change from fill to cut
☐ c) a surplus
☐ d) a balance of cut and fill

64. _____ is a type of intersection where the major road passes over or under the minor road using a bridge, overpass, or underpass. * 1 point

Mark only one oval.

- ☐ a) Round about
☐ b) Channelization
☐ c) Interchange
☐ d) at-grade separation

57. Which of the following strategies can help mitigate traffic congestion? *

Mark only one oval.

- ☐ a) Reducing the number of traffic signals
- ☐ b) Increasing the number of lanes on the highway
- ☐ c) Encouraging more people to use public transportation
- ☐ d) All of the above

58. Which of the following is NOT a characteristic of LOS-C on a basic freeway segment?

1 point

Mark only one oval.

- ☐ a) Drivers have reasonable freedom to maneuver
- ☐ b) Traffic flow is stable, but beginning to become affected by the presence of other vehicles.
- ☐ c) Drivers may experience some delays and still considered acceptable levels of congestion.
- ☐ d) Traffic flow is stable, and it is unaffected by the presence of other vehicles.

59. _____ is not basic stages of the highway planning and development process for feasibility stage?

* 1 point

Mark only one oval.

- ☐ a) Assess the need for a new highway or highway improvements
- ☐ b) Evaluate potential corridors and identify constraints (environmental, social, economic)
- ☐ c) Perform preliminary cost-benefit analysis
- ☐ d) Analyze factors such as terrain, soil conditions, drainage, and constructability

60. _____ is a detail study surveying route selection stage, if more than one routes passed the route selection stage? *

1 point

Mark only one oval.

- ☐ a) Reconnaissance
- ☐ b) Desk study
- ☐ c) Preliminary Survey
- ☐ d) Route location Survey



53. A study of freeway flow at a particular site has resulted in a calibrated speed-density relationship, as follows:

* 1 point

$U_s = 57.5(1 - 0.008k)$. From this relation, Find the free-flow speed and jam density?

Mark only one oval.

- ☐ a) 57.5 Km/hr, 126 veh/lane.
- ☐ b) 57.5 Km/hr, 126veh/km.
- ☐ c) 28.75 Km, hr./ 120veh/km
- ☐ d) 115 Km/hr, 120 veh/ lane

54. _____ is a traffic sign indicating the Speed Limit on a road?*

1 point

Mark only one oval.

- ☐ (A) Circular Shape with white Background and Red Border
- ☐ (B) Triangular Shape with White Background and Red Border
- ☐ (C) Triangular Shape with Red Background and White Border
- ☐ (D) Circular Shape with Red Background and White Border

55. _____ not the general criteria used in route selection *

1 point

Mark only one oval.

- ☐ a) Alignment maximizes for land acquisition
- ☐ b) The relative length of the alternatives was considered
- ☐ c) Routes result in least environmental disturbance is preferred.
- ☐ d) Route with least overall project cost are preferred.

56. From the speed-density relationship what is the current speed (u), if $u_f = 90$ km/h and $k_j = 180$ vehicles/km, and the current density is 90 vehicles/km?

* 1 point

Mark only one oval.

- ☐ a) 90 km/h
- ☐ b) 72 km/h
- ☐ c) 45 km/h
- ☐ d) 18 km/h

65. _____ is the layer of pavement material that provides a means for water to drain away from the pavement structure. * 1 point



Mark only one oval.

- ☐ a) Surface course
- ☐ b) Subbase
- ☐ c) Subgrade
- ☐ d) Base course

66. Compared to rigid pavements, flexible pavements generally require _____ maintenance. * 1 point

Mark only one oval.

- ☐ a) More
- ☐ b) Less
- ☐ c) The same amount of
- ☐ d) No

67. A bitumen concrete mix has average specific gravity of 2.325 and theoretical specific gravity as 2.41. The density of bitumen is 1.03 g/cc with 4.5% bitumen content by weight in mix. The VFB of mix is * 1 point

Mark only one oval.

- ☐ a) 32.8%
- ☐ b) 30.8%
- ☐ c) 20.73%
- ☐ d) 10.15%

68. In Marshall method of mix design, the coarse aggregates, fine aggregates, filler material and bitumen, having respective specific gravities of 2.82, 2.42, 2.68, and 1.02 are mixed in ratio of 52, 34.8, 4.2 and 5.4 percent respectively. The theoretical specific gravity of the mix would be ____ * 1 point

Mark only one oval.

- ☐ a) 2.12
- ☐ b) 2.52
- ☐ c) 2.04
- ☐ d) 2.25

69. Which of the following factors is the LEAST important in choosing a stabilization technique for a pavement structure? * 1 point

Mark only one oval.

- ☐ A. Type of pavement (flexible, rigid, or composite)
- ☐ B. Traffic loads
- ☐ C. Environmental conditions
- ☐ D. Availability of local materials

70. For a heavily trafficked flexible pavement, which stabilization technique would be the most suitable? * 1 point

Mark only one oval.

- ☐ A. Mechanical stabilization using geosynthetics
- ☐ B. Chemical stabilization with cement
- ☐ C. Bituminous stabilization with asphalt emulsion
- ☐ D. Cement concrete stabilization

71. Why is it important to assess the environmental factors that can affect the behavior and durability of construction materials? * 1 point

Mark only one oval.

- ☐ A. To save costs on construction materials
- ☐ B. To ensure compliance with building codes
- ☐ C. To minimize the risk of structural failure
- ☐ D. To improve the aesthetics of the building

72. How does increasing the water content in the mix impact the strength of concrete? * 1 point

Mark only one oval.

- ☐ A. It increases the strength of concrete
- ☐ B. It decreases the strength of concrete
- ☐ C. It has no impact on the strength of concrete
- ☐ D. It only impacts the workability of the mix

73. What is the purpose of a slump test for concrete? *

Mark only one oval.

- ☐ A. To determine the compressive strength of the concrete
- ☐ B. To evaluate the durability of the concrete
- ☐ C. To assess the workability of the concrete mix
- ☐ D. To test the permeability of the concrete



74. How does the moisture content of timber affect its use in construction? *

1 point

Mark only one oval.

- ☐ A. High moisture content can make the wood more difficult to work with
- ☐ B. Low moisture content can make the wood more fragile and susceptible to damage
- ☐ C. Moisture content affects the dimensional stability of the wood over time
- ☐ D. All of the above

75. What is the primary benefit of using clay products instead of concrete in Ethiopia's construction industry? *

1 point

Mark only one oval.

- ☐ A. They are more weather-resistant and durable
- ☐ B. They provide better thermal insulation
- ☐ C. They are more environmentally friendly
- ☐ D. None of the above

76. Which test measures the strength and durability of building stone under compressive forces?

* 1 point

Mark only one oval.

- ☐ A. Tensile strength test
- ☐ B. Water absorption test
- ☐ C. Acid test
- ☐ D. Compressive strength test

77. What is a common practice to streamline the construction process in cost-efficient construction? * 1 point



Mark only one oval.

- ☐ A) Following traditional construction methods
- ☐ B) Using only custom-made materials
- ☐ C) Minimizing waste and maximizing efficiency
- ☐ D) Implementing costly and complex construction techniques

78. Which of the following is typically included in a site plan? * 1 point

Mark only one oval.

- ☐ A) Property boundaries and setbacks
- ☐ B) Interior wall colors
- ☐ B) Structural engineering details
- ☐ D) HVAC system specifications

79. What is the purpose of using prefabrication in building construction? * 1 point

Mark only one oval.

- ☐ A) To increase labor costs
- ☐ B) To reduce construction time
- ☐ C) To maximize on-site customization
- ☐ D) To minimize material usage

80. Which construction technique involves the use of environmentally sustainable materials and energy-efficient design principles? * 1 point

Mark only one oval.

- ☐ A) Traditional construction
- ☐ B) Green building construction
- ☐ C) Concrete construction
- ☐ D) Steel frame construction

81. When designing a building, how does the placement of columns and walls impact the overall architectural layout? * 1 point



Mark only one oval.

- ☐ A) Walls can be easily relocated, while columns are fixed in place
- ☐ B) Columns can be used as decorative elements, while walls are purely functional
- ☐ C) Walls determine the structural stability, while columns control natural ventilation
- ☐ D) Columns define the spatial organization, while walls serve as structural boundaries

82. Which technology is commonly used for creating detailed 3D models of building designs before construction begins? * 1 point

Mark only one oval.

- ☐ A) Building Information Modeling (BIM)
- ☐ B) Augmented Reality (AR)
- ☐ C) Virtual Reality (VR)
- ☐ D) Computer-Aided Design (CAD)

83. What is the net section area of steel plate 40cm wide and 10mm thick with one bolt if diameter of bolt hole is 18mm? * 1 point

Mark only one oval.

- ☐ a) 38.2 cm²
- ☐ b) 24 cm²
- ☐ c) 578 mm²
- ☐ d) 465 mm²

84. Which of the following factor is considered for classification of cross section? * 1 point

Mark only one oval.

- ☐ a) length of member
- ☐ b) location where member is used
- ☐ c) seismic force
- ☐ d) width-to-thickness ratio

85. The design tensile strength of tensile member is *

Mark only one oval.

- ☐ a) minimum of strength due to gross yielding, net section rupture, block shear
- ☐ b) maximum of strength due to gross yielding, net section rupture, block shear
- ☐ c) strength due to gross yielding
- ☐ d) strength due to block shear

86. In Working Stress Method, which of the following relation is correct? *

1 point

Mark only one oval.

- ☐ a) Working Stress $\leq$ Permissible Stress
- ☐ b) Working Stress $\geq$ Permissible Stress
- ☐ c) Working Stress = Permissible Stress
- ☐ d) Working Stress $>$ Permissible Stress

87. What is the yield strength of bolt of class 4.6? *

1 point

Mark only one oval.

- ☐ a) 400 N/mm²
- ☐ b) 240 N/mm²
- ☐ c) 250 N/mm²
- ☐ d) 500 N/mm²

88. For very short compression member *

2 points

Mark only one oval.

- ☐ a) failure stress will be greater than yield stress
- ☐ b) failure stress will be less than yield stress
- ☐ c) failure stress will equal yield stress
- ☐ d) failure stress will be twice the yield stress



89. An ideal site for bridge crossing is one that *

Mark only one oval.

- ☐ a) is on a straight reach of the river
- ☐ b) has well defined and stable high banks above flood level
- ☐ c) does not require expensive river training work
- ☐ d) all

90. Design vehicular live loads on roadways of bridges consist of the combination of* 1 point

Mark only one oval.

- ☐ a) Design truck and design lane load
- ☐ b) Design tandem and Design lane load
- ☐ c) $1.25DC + 1.5DW + 1.75LL$
- ☐ d) a and b

91. Due to hammering effect of wheel assembly on riding surface the live load effects * 1 point shall be increased. The factor which accounts this is

Mark only one oval.

- ☐ a) Dynamic load allowance
- ☐ b) Multiple presence factor
- ☐ c) Distribution factor
- ☐ d) Load factor

92. Determine the number of design lane if the clear roadway is 9m.*

1 point

Mark only one oval.

- ☐ a) 2
- ☐ b) 1
- ☐ c) 3
- ☐ d) 4

94. Which statement is not correct about service limit state *

33 / 35

Mark only one oval.

- ☐ a) It includes deformation and crack control
- ☐ b) The load modifier is one
- ☐ c) The spacing between middle axle and rare axle shall be 9m for investigation of load effects.
- ☐ d) All



95. In yield line theory, yield lines typically form at locations where: *

1 point

Mark only one oval.

- ☐ a) The bending moment is zero
- ☐ b) The shear force is maximum
- ☐ c) The slab experiences the highest deflection
- ☐ d) The moment capacity of the slab is exceeded

96. A square slab 5m x 5m is isotropically reinforced at the bottom. If it is subjected to a design load of 12 kN/m² the moment capacity required as yield line theory is *

2 points

Mark only one oval.

- ☐ a) 6.25 kN.m
- ☐ b) 18 kN.m
- ☐ c) 25 kN.m
- ☐ d) 12.5 kN.m

97. How should the reinforcement be handled around a significant opening in a slab designed with the strip method?

* 1 point

Mark only one oval.

- ☐ a) Reinforcement should be decreased near the opening to save material
- ☐ b) Additional reinforcement should be provided around the edges of the opening
- ☐ c) Reinforcement should be eliminated around the opening to allow for more flexibility
- ☐ d) Reinforcement should be designed without any additional consideration



98. Which element is typically designed to yield first in a capacity-based design approach?

* 1 point

Mark only one oval.

- ☐ a) Columns
- ☐ b) Beams
- ☐ c) Both columns and beams have equal strength
- ☐ d) Yield sequence is not a problem

99. A fixed beam with 4m span length is subjected to a concentrated load of 20kN at the mid length of the beam. What will be the required value of M_p to ensure a minimum load factor $\lambda=1.7$.

* 2 points

Mark only one oval.

- ☐ a) 68kN.m
- ☐ b). 34 kN.m
- ☐ c) 17kN.m
- ☐ d). 10kN.m

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